

Auteur: Jeremi Lorek  
Studierichting: Informatica  
Oefeningenreeks 4A nr. 40.4

$$\begin{aligned} & \int \frac{dx}{\sqrt{-3 + 4x - x^2}} \\ &= \int \frac{dx}{\sqrt{-(x^2 - 4x + 3)}} \\ &= \int \frac{dx}{\sqrt{-(x^2 - 4x + 4 - 1)}} \\ &= \int \frac{dx}{\sqrt{-((x - 2)^2 - 1)}} \\ &= \int \frac{dx}{\sqrt{1 - (x - 2)^2}} \end{aligned}$$

Stel  $u = x - 2 \Rightarrow du = dx$

$$\begin{aligned} &= \int \frac{du}{\sqrt{1 - u^2}} \\ &= \text{Bgsin } u + C = \text{Bgsin}(x - 2) + C \end{aligned}$$

## References